

Eightfold Assignment

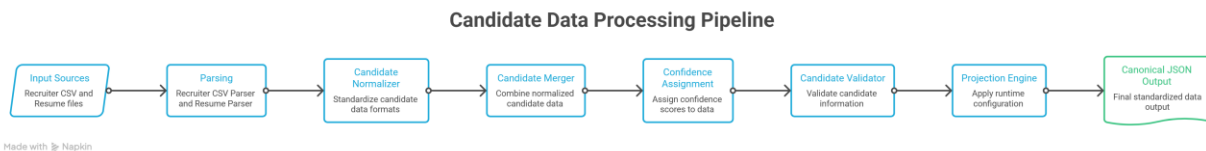
Multi-Source Candidate Data Transformer

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The Multi-Source Candidate Data Transformer integrates candidate information from Recruiter CSV and resume files into a unified profile by applying normalization, conflict resolution, validation, and configurable JSON generation

Canonical Output Schema & Normalization:

The system converts data from multiple sources into a single canonical candidate profile. A common schema is used for every candidate to maintain consistency across different inputs. Email addresses are converted to lowercase, and duplicate entries are removed. Phone numbers are normalized in the E.164 format for uniform representation. Skills are stored using a consistent naming format with duplicates eliminated. Education and experience details are extracted from the resume and organized into structured objects. Provenance information is maintained for each field to record its source, and an overall confidence score is included to indicate the reliability of the final profile.



Merge Policy & Confidence Assignment:

Candidate records from the Recruiter CSV and Resume are merged into one profile using a priority-based approach. The candidate's name is primarily taken from the Recruiter CSV, while skills, education, and experience are extracted from the resume. Email addresses and phone numbers from both sources are normalized before merging to avoid duplicates. When conflicting values exist, the system keeps the most reliable information based on the source. A confidence score is calculated from the completeness of the merged profile, providing an overall measure of data quality.

Runtime Configuration (Projection & Validation):

The output is generated through a configurable projection layer. The Projection Engine reads the runtime configuration from config.json and includes only the selected fields in the final JSON without requiring code changes. Before writing the output, the Candidate Validator verifies that mandatory fields follow the expected schema and that invalid or incomplete data does not affect the pipeline execution.

Edge Cases Handled:

- Missing email or phone values are safely ignored without interrupting processing.
- Duplicate email addresses and phone numbers are removed after normalization.
- Resume files in TXT, PDF, and DOCX formats are automatically detected and parsed.
- Common PDF text extraction issues are handled before processing the content.
- If optional information is unavailable, the corresponding fields remain empty instead of using incorrect values.

Scope Left for Future Work:

Due to time constraints, some advanced features were intentionally kept out of the current implementation. These include LinkedIn profile integration, GitHub profile parsing, OCR support for scanned resumes, and processing multiple candidates in a single execution. The current design is modular, making it straightforward to add these capabilities in future versions.